

GDI nonlinear equations

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1. Basic equations

Continuity, ion momentum, electron momentum, quasineutrality. Ion velocity and number density are $\mathbf{u}$ and n respectively to avoid having to write subscripts constantly. Cold ions (no ITG).

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\mathbf{u}) = 0 \quad (1.1)$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \mathbf{u}_e) = 0 \quad (1.2)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = \frac{Ze}{m_i} \left[\mathbf{E} + \frac{\mathbf{u} \times \mathbf{B}}{c} \right] - \nu_{in} \mathbf{u} \quad (1.3)$$

$$\frac{\partial \mathbf{u}_e}{\partial t} + \mathbf{u}_e \cdot \nabla \mathbf{u}_e = -\frac{e}{m_e} \left[\mathbf{E} + \frac{\mathbf{u}_e \times \mathbf{B}}{c} \right] - \frac{T_e}{nm_e} \nabla n - \nu_{en} \mathbf{u}_e \quad (1.4)$$

$$n_e = Zn \quad (1.5)$$

2. GDI physics: 2D

2.1. Orderings and equilibrium

We assume electrostatic field $E = E_0 \hat{\mathbf{y}} - \nabla \phi$ (this is essentially a requirement on the frequency, lets work it out afterwards). We also assume $n = n_0(x - v_0 t) + \delta n$, where $v_0 = cE_0/B$. We assume $n_0(x) \gg \delta n$ and $E_0 \gg k_\perp \phi$, but that $L_n = n_0/(dn_0/dx) \gg 1/k_\perp (\gg \rho_s)$, such that $k_\perp \delta n \sim n_0/L_n$. We also assume $\omega, \nu_{in} \ll \Omega_i$, so we can use the drift approximation.

Finally, we assume that $\partial/\partial z = 0$. We will come back to this later. Note that in this case, there is no way for the electrons to become adiabatic.

2.2. Electrons

The electron drift is ($\mathbf{B} \times$ momentum equation, neglect terms small by order of the electron gyrofrequency, solve)

$$\mathbf{u}_{e\perp} = \frac{c\mathbf{E} \times \mathbf{B}}{B^2} - \frac{cT_e}{enB^2} \mathbf{B} \times \nabla n. \quad (2.1)$$

Plugging this into the electron continuity equation, and writing it in the frame moving with the velocity $\mathbf{v}_0 = \hat{\mathbf{x}}cE_0/B$, we find

$$\frac{\partial}{\partial t} \left(\frac{\delta n}{n_0} \right) - \frac{c}{B} \frac{1}{L_n} \frac{\partial \phi}{\partial y} + \frac{c}{B} \left\{ \phi, \frac{\delta n}{n_0} \right\} = 0, \quad (2.2)$$

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where we have used the separation of scales between L_n and $1/k_\perp$ to divide through by the average of n_0 over the small fluctuations. The term involving L_n is the one needed for the GDI. This is the first of our equations.

2.3. Ions

The expression for the ion drift is, using the momentum equation and solving iteratively (drift approx) to first order,

$$\mathbf{u}_\perp = \mathbf{v}_0 + \frac{c}{B_0} \hat{\mathbf{z}} \times \nabla_\perp \phi - \frac{c}{B_0 \Omega_i} [d_t + \nu_{in}] \nabla_\perp \phi + \frac{c \nu_{in}}{B_0 \Omega_i} \mathbf{E}_{\perp 0}. \quad (2.3)$$

Included here are the $E \times B$, polarization, and Pedersen drifts. We subtract the ion continuity equation from the electron continuity equation, leading to an expression for $\nabla \cdot \mathbf{J} = 0$,

$$\nabla_\perp \cdot [n(d_t + \nu_{in}) \nabla_\perp \phi] - \nu_{in} E_0 \frac{\partial \delta n}{\partial y} = 0. \quad (2.4)$$

Expanding out the first term, doing a little algebra, and dropping all but the leading order terms, we find

$$\frac{\partial \nabla_\perp^2 \phi}{\partial t} + \frac{c}{B} \{\phi, \nabla_\perp^2 \phi\} + \nu_{in} \nabla_\perp^2 \phi - \nu_{in} E_0 \frac{\partial}{\partial y} \left(\frac{\delta n}{n_0} \right) = 0. \quad (2.5)$$

The final term is the GDI drive term, coupling the evolution of the vorticity and the density. This is the second equation.

2.4. Nonlinear equations: summary

To clean this up, we write $\Phi = c\phi/B$, $v_0 = cE_0/B$.

$$\frac{\partial}{\partial t} \left(\frac{\delta n}{n_0} \right) - \frac{1}{L_n} \frac{\partial \Phi}{\partial y} + \left\{ \Phi, \frac{\delta n}{n_0} \right\} = 0, \quad (2.6)$$

$$\frac{\partial \nabla_\perp^2 \Phi}{\partial t} + \{\Phi, \nabla_\perp^2 \Phi\} + \nu_{in} \nabla_\perp^2 \Phi - \nu_{in} v_0 \frac{\partial}{\partial y} \left(\frac{\delta n}{n_0} \right) = 0. \quad (2.7)$$

2.5. Linear dispersion relation

Linearizing, we find

$$\omega(\omega + i\nu_{in}) = -\frac{k_y^2}{k_\perp^2} \frac{\nu_{in} v_0}{L_n} = -A \quad (2.8)$$

defining the drive parameter A , which has units of frequency squared. For $A \ll \nu_{in}^2/2$, we have the collisional GDI growth rate,

$$\omega = i \frac{A}{\nu_{in}}, -i(\nu_{in} + A/\nu_{in}), \quad (2.9)$$

so the growing mode has

$$\gamma = \frac{k_y^2}{k_\perp^2} \frac{v_0}{L_n}, \quad (2.10)$$

including the dependence on $k_y/k_\perp$.

For $A \gg \nu_{in}^2/2$, we have the Keskinen & Ossakow version including the ion inertia (effectively, the ion polarization drift),

$$\omega = \pm i \sqrt{A} = \pm i \frac{k_y}{k_\perp} \sqrt{\frac{\nu_{in} v_0}{L_n}}, \quad (2.11)$$

at lowest order.

2.6. Relationship between density and potential

We can read off the (linearized) relationship between the density and potential from Eq. (2.6):

$$\frac{\delta n}{n_0} = -\frac{k_y}{\omega L_n} \Phi. \quad (2.12)$$

This is very different from adiabatic: this difference is in fact what allows the instability. Consider what happens if you plug the adiabatic relationship, $\Phi \propto \delta n/n_0$, into Eq. (2.7): then the equations decouple, and due to the lack of phase difference between δn and Φ , there is no growth.

2.7. Saturation

The instability growth rate is maximized for $k_\perp$ in the y direction, i.e. long fingers aligned with the background gradient. Saturation requires nonlinear interaction, i.e. $k_\perp$ not totally aligned. One idea for how this occurs: secondary instability of the fingers via KH. This has a maximum growth rate of order U/L_y , with maximum growth at $k_x \sim 1/L_y$, isotropizing the fingers. This then leads to a saturated amplitude

$$k_y^2 \Phi_{sat} \sim \gamma_{max} \quad (2.13)$$

This leads to an electric field (or $E \times B$) spectrum $\propto k_\perp^{-3}$. The density spectrum should follow this, see earlier. However, we have not yet considered the cascade of the turbulence.

2.8. Cascade

The saturation spectrum in fact coincides with the classic direct cascade of enstrophy. Along with this there could potentially be an inverse cascade, but since the 2D hydro inverse cascade is a $-5/3$ spectrum, this will likely be swamped by the saturation spectrum for the GDI. This is a fundamentally large-scale instability. THIS ALL SHOULD BE SIMULATED!

2.9. Keton turbulence

The fact that we are seeing an adiabatic relationship here suggests that this 2D theory breaks down. To handle this, we need to include some parallel gradients in the problem, allowing the adiabatic relationship to be set up. There may be some transition to drift-wave turbulence at the scales that Keton is probing.

3. Three-dimensional model

Here we include the parallel diffusivity. This will turn out to regularize the large-scale instability, while also introducing drift wave dynamics.

3.1. Orderings

We order $\Omega_e \gg \nu_{en} \gg \omega$, i.e. we are going to neglect electron inertia (is this justifiable?). We also order

$$\left(\frac{k_\parallel}{k_\perp}\right)^2 \sim \epsilon \frac{m_e}{m_i} \frac{\nu_{en}}{\Omega_i} \quad (3.1)$$

in addition to our earlier orderings. This is to make the linearized parallel electron current comparable to the other terms in the continuity equations.

3.2. Electrons

In the perpendicular direction we have the same electron perpendicular drifts (2.1). The parallel electron velocity is given, to lowest order, by

$$u_{e\parallel} = \frac{e}{m_e \nu_{en}} \left[\frac{\partial \phi}{\partial z} - \frac{T_e}{e} \frac{1}{n} \frac{\partial n}{\partial z} \right]. \quad (3.2)$$

We must add this to our electron continuity equation, yielding

$$\frac{\partial}{\partial t} \left(\frac{\delta n}{n_0} \right) - \frac{c}{B} \frac{1}{L_n} \frac{\partial \phi}{\partial y} + \frac{c}{B} \left\{ \phi, \frac{\delta n}{n_0} \right\} + \frac{e}{m_e \nu_{en}} \frac{\partial^2}{\partial z^2} \left[\phi - \frac{T_e}{e} \left(\frac{\delta n}{n_0} \right) \right] = 0, \quad (3.3)$$

3.3. Ions

There is no change to the ion dynamics, since the ion parallel current is small by order of the mass ratio. Subtracting the ion continuity equation from the electron continuity equation above, and dropping all but leading-order terms,

$$\frac{\partial \nabla_{\perp}^2 \phi}{\partial t} + \frac{c}{B} \{ \phi, \nabla_{\perp}^2 \phi \} + \nu_{in} \nabla_{\perp}^2 \phi - \nu_{in} E_0 \frac{\partial}{\partial y} \left(\frac{\delta n}{n_0} \right) = - \frac{\Omega_i \Omega_e}{\nu_{en}} \frac{\partial^2}{\partial z^2} \left[\phi - \frac{T_e}{e} \frac{\delta n}{n_0} \right]. \quad (3.4)$$

3.4. 3D nonlinear equations: summary

This time it is more convenient to normalize $\varphi = e\phi/T_e$, $N = \delta n/n_0$, $v_0 = cE_0/B$, and define the parallel diffusivity $D_{\parallel} = \Omega_i \Omega_e \rho_s^2 / \nu_{en} = v_{the}^2 / \nu_{en}$.

$$\frac{\partial N}{\partial t} - \frac{\rho_s c_s}{L_n} \frac{\partial \varphi}{\partial y} + \rho_s c_s \{ \varphi, N \} + D_{\parallel} \frac{\partial^2}{\partial z^2} [\varphi - N] = 0, \quad (3.5)$$

$$\frac{\partial \nabla_{\perp}^2 \varphi}{\partial t} + \rho_s c_s \{ \varphi, \nabla_{\perp}^2 \varphi \} + \nu_{in} \nabla_{\perp}^2 \varphi - \frac{\nu_{in} v_0}{\rho_s c_s} \frac{\partial N}{\partial y} + \frac{D_{\parallel}}{\rho_s^2} \frac{\partial^2}{\partial z^2} [\varphi - N] = 0. \quad (3.6)$$

3.5. Linear instability

Let's also write characteristic frequencies $\omega_* = k_y \rho_s c_s / L_n$, $\omega_{Ped} = k_y \nu_{in} v_0 / \Omega_i$, $\gamma_{\parallel} = D_{\parallel} k_z^2$. Linearizing,

$$(\omega + i\gamma_{\parallel}) ([\omega + i\nu_{in}] k_{\perp}^2 \rho_s^2 + i\gamma_{\parallel}) = -(\omega_{Ped} + i\gamma_{\parallel}) (\omega_* - i\gamma_{\parallel}). \quad (3.7)$$

From the first equation we also have

$$N = \left(\frac{\gamma_{\parallel} + i\omega_*}{\gamma_{\parallel} - i\omega} \right) \varphi, \quad (3.8)$$

from which we can see how large $\gamma_{\parallel}$ pulls the electrons into the Boltzmann relation. In the following we will study this dispersion relation in various analytically tractable limits.

3.6. 2D limit

Noting that $\omega_{Ped} \omega_* = A k_{\perp}^2 \rho_s^2$, as $\gamma_{\parallel} \rightarrow 0$ we recover the 2D dispersion relation, as expected.

3.7. Nearly-adiabatic limit

As $\gamma_{\parallel} \rightarrow \infty$, we write $\omega = \omega_0 + \delta\omega$ where $\delta\omega/\omega_0 \sim \omega_0/\gamma_{\parallel} \ll 1$. At lowest order, we get drift waves modified by the Pedersen drift and collisional damping:

$$\omega_0 = \frac{\omega_{Ped} - \omega_* - i\nu_{in} k_{\perp}^2 \rho_s^2}{1 + k_{\perp}^2 \rho_s^2} = \frac{k_{\perp} \rho_s \left(\frac{k_y}{k_{\perp}} \right) \left(\frac{\nu_{in} v_0}{c_s} - \frac{c_s}{L_n} \right) - i\nu_{in} k_{\perp}^2 \rho_s^2}{1 + k_{\perp}^2 \rho_s^2}, \quad (3.9)$$

where we have written the second bracket to make the overall dependence on $k_\perp \rho_s$ clearer.

3.8. General behaviour

We are interested in the behaviour as a function of $k_\perp \rho_s$ when all timescales are comparable. Physically, we expect the position of the maximum growth rate to move to larger scales as $\gamma_\parallel$ decreases, while smoothly increasing to the 2D GDI limit and becoming independent of $\gamma_\parallel$ as $\gamma_\parallel \rightarrow 0$. To tackle this case, we write $\omega = \omega_R + i\gamma$. The imaginary part of the dispersion relation yields

$$\omega_R = \frac{\gamma_\parallel (\omega_{\text{Ped}} - \omega_*)}{(2\gamma + \nu_{in})k_\perp^2 \rho_s^2 + \gamma_\parallel (1 + k_\perp^2 \rho_s^2)}. \quad (3.10)$$

Inserting this into the real part of the dispersion relation yields a quartic for the growth/damping rate γ :

$$\gamma^2 + \gamma \left[\nu_{in} + \gamma_\parallel \frac{1 + k_\perp^2 \rho_s^2}{k_\perp^2 \rho_s^2} \right] - [A - \gamma_\parallel \nu_{in}] = \left[\frac{\gamma_\parallel (\omega_* - \omega_{\text{Ped}})}{(2\gamma + \nu_{in})k_\perp^2 \rho_s^2 + \gamma_\parallel (1 + k_\perp^2 \rho_s^2)} \right]^2. \quad (3.11)$$

The final term in square brackets on the LHS has a piece corresponding to the GDI drive A , and another piece corresponding to collisional damping. The RHS corresponds to drift waves driven unstable by the parallel diffusivity (à la Hasegawa-Wakatani); indeed, setting $\nu_{in} = 0$ (which also sets $A = 0$), we recover for $\gamma \ll \gamma_\parallel$,

$$\gamma = \frac{\omega_*^2 k_\perp^2 \rho_s^2}{\gamma_\parallel (1 + k_\perp^2 \rho_s^2)^3}, \quad (3.12)$$

the standard Hasegawa-Wakatani growth rate. Restoring $\nu_{in} > 0$, as $k_\perp \rho_s \rightarrow 0$, the growth rate vanishes as $k_\perp^2 \rho_s^2$:

$$\gamma \approx \frac{A + \left[\frac{\gamma_\parallel (\omega_* - \omega_{\text{Ped}})}{\nu_{in} k_\perp^2 \rho_s^2 + \gamma_\parallel} \right]^2 - \gamma_\parallel \nu_{in}}{\nu_{in} k_\perp^2 \rho_s^2 + \gamma_\parallel} k_\perp^2 \rho_s^2, \quad k_\perp^2 \rho_s^2 \ll 1 \quad (3.13)$$

The factor with $\omega_* - \omega_{\text{Ped}}$ (the H-W drive) will eventually become irrelevant at large enough scales, since it is proportional to k_y^2 , leaving just the GDI drive A .

At small scales, while the Hasegawa-Wakatani growth rate decreases rapidly, the GDI part asymptotes to a $k_\perp \rho_s$ -independent rate,

$$\gamma \approx -\frac{\nu_{in} + \gamma_\parallel}{2} \pm \frac{1}{2} \sqrt{(\nu_{in} + \gamma_\parallel)^2 + 4(A - \gamma_\parallel \nu_{in})}, \quad k_\perp^2 \rho_s^2 \gg 1 \quad (3.14)$$

unstable provided that $A > \gamma_\parallel \nu_{in}$. In reality, this limit is fairly suspect given that effects like ion gyroviscosity have not been included, which will add a additional $k_\perp$ -dependent damping term to our equations. Finally, we note that the mode is stabilized if

$$A + \left[\frac{\gamma_\parallel (\omega_* - \omega_{\text{Ped}})}{\nu_{in} k_\perp^2 \rho_s^2 + \gamma_\parallel (1 + k_\perp^2 \rho_s^2)} \right]^2 < \gamma_\parallel \nu_{in}. \quad (3.15)$$

Since $\gamma_\parallel \propto k_z^2$, high- k_z modes are damped.

3.9. Conserved quantities

Let us examine the conserved quantities in the system obtained setting $v_0 = 0$, $1/L_n = 0$ (i.e. no instabilities). The energy is given by

$$E = \frac{1}{2} \int d^3 \mathbf{x} [N^2 + \rho_s^2 |\nabla_\perp \varphi|^2], \quad (3.16)$$

while the potential enstrophy is given by

$$Z = \frac{1}{2} \int d^3 \mathbf{x} [N - \rho_s^2 \nabla_\perp^2 \varphi]^2. \quad (3.17)$$

The energy is strictly dissipated by the diffusivity and ion-neutral collisions:

$$\frac{dE}{dt} = - \int d^3 \mathbf{x} [\nu_{in} \rho_s^2 |\nabla_\perp \varphi|^2 + D_\parallel (\partial_z (\varphi - N))^2]. \quad (3.18)$$

On the other hand, the evolution of potential enstrophy is not necessarily negative definite, provided $\nu_{in} \neq 0$:

$$\frac{dZ}{dt} = \nu_{in} \int d^3 \mathbf{x} [N \rho_s^2 \nabla_\perp^2 \varphi - (\rho_s^2 \nabla_\perp^2 \varphi)^2]. \quad (3.19)$$

In the adiabatic limit ($\gamma_\parallel \rightarrow \infty$ in Eq. 3.8), $N \sim \varphi$, making the whole expression negative definite. In the opposite (2D GDI) limit, $\gamma_\parallel \rightarrow 0$, linearly, ω is purely imaginary, so that the first term is approximately zero. Nonlinearly, this is an open question...

4. Cutoff scale and predictions for density and electric field spectra.

Lets work in the low-frequency regime, i.e. $\gamma, \omega \ll \nu_{in}$. Balancing terms in Eq. (3.6), we have

$$\nu_{in} \nabla_\perp^2 \varphi \sim \frac{D_\parallel}{\rho_s^2} \partial_z^2 \varphi. \quad (4.1)$$

This essentially provides a minimum $k_\parallel$ for a given $k_\perp$, required to maintain current closure (I think you all call this mapping in the ionospheric community?),

$$k_\parallel^{\min} = \sqrt{\nu_{in}/D_\parallel} k_\perp \rho_s. \quad (4.2)$$

This is only valid provided that $k_\parallel^{\min} L_F \gg 1$: at larger scales, one has to take into account current closure in the E region. Alternatively, we can concisely write this as

$$\gamma_\parallel^{\min} = \nu_{in} k_\perp^2 \rho_s^2. \quad (4.3)$$

Plugging this into Eq. (3.13), valid at $k_\perp^2 \rho_s^2 \ll 1$, and neglecting the factor $(\omega_* - \omega_{\text{Ped}})^2$ (this is an assumption that the cutoff scale derived below is large compared to ρ_s), we obtain

$$\gamma = \frac{1}{2} \left[\frac{A}{\nu_{in}} - \nu_{in} k_\perp^2 \rho_s^2 \right], \quad (4.4)$$

which defines a critical scale for marginal stability,

$$k_\perp^c = \frac{\sqrt{A}}{\nu_{in} \rho_s} = \frac{1}{\rho_s} \left| \frac{k_y}{k_\perp} \right| \sqrt{\frac{v_0}{\nu_{in} L_n}} \quad (4.5)$$

This argument could be formalized by height-integrating the equations over z . It could also be generalized to include the ion inertia: a critical-balance-like argument, as Tamar suggested.

What should we expect either side of this break? For $k \ll k_\perp^c$, we expect the GDI relationship Eq. (2.12), which, writing out the lowest-order expressions for ω (with the minimum $\gamma_\parallel$, i.e. given by Eq. 4.4 above) and writing everything in terms of N and φ , becomes

$$N = i k_y \frac{k_\perp^2}{k_y^2} \frac{2 \rho_s c_s}{v_0} \varphi. \quad (4.6)$$

This predicts that the density and electric field $E = -i\mathbf{k}\phi$ spectra should be 180° out of phase, with a specific (and constant with $|k_\perp|$) ratio between them. The factor 2 in the above equation relates to the floor in $\gamma_\parallel$: if we instead assumed $k_\parallel = 0$ (exactly 2D), it would be 1 instead. This could instead be parametrized as some value α : a test of the effective parallel scale:

$$N = ik_y \frac{k_\perp^2}{k_y^2} \frac{\alpha \rho_s c_s}{v_0} \varphi, \quad \alpha = 1 + D_\parallel k_z^2 / \nu_{in} k_\perp^2 \rho_s^2. \quad (4.7)$$

Conversely, for $k \gg k_\perp^c$, we expect the adiabatic relationship,

$$N = \varphi, \quad (4.8)$$

implying density and electric field 90° out of phase, and the standard adiabatic relationship between them, including a factor k , meaning a factor k^2 between the spectra. Encouragingly, Kenton's results show this behavior on both sides of the break quite clearly (with the check on the ratio at large scales pending).

We currently do not have a detailed explanation for the actual value of the spectral index on either side of the break: but at large scales, the density appears to look something between -2 and -3 (maybe?), similarish to the theory earlier in these notes. If we assume this continues into the drift-wave regime, then the electric field flattening is implied by the adiabatic relationship. However we are not necessarily sure about the exact mechanism for the drift wave driving: maybe secondary instability of the GDI fingers, or driving via resistive drift instability at $k_\perp \rho_s \sim 1$?

5. Numerical implementation

I will implement this in the `jax_rhhd` framework. Lets assume we have a fixed k_z ; transforming the equations, we have

$$\frac{\partial \tilde{N}}{\partial t} - \frac{\rho_s c_s}{L_n} ik_y \tilde{\varphi} + \rho_s c_s \{\widetilde{\varphi, N}\} - \gamma_\parallel [\tilde{\varphi} - \tilde{N}] = 0, \quad (5.1)$$

$$\frac{\partial \tilde{\varphi}}{\partial t} - \frac{\rho_s c_s}{k_\perp^2} \{\widetilde{\varphi, \nabla_\perp^2 \varphi}\} + \nu_{in} \tilde{\varphi} + \frac{\nu_{in} v_0}{k_\perp^2 \rho_s c_s} ik_y \tilde{N} + \frac{\gamma_\parallel}{k_\perp^2 \rho_s^2} [\tilde{\varphi} - \tilde{N}] = 0 \quad (5.2)$$

It will be convenient to solve the linear terms exactly via integrating factor: these amount to a matrix

$$\frac{\partial}{\partial t} \begin{bmatrix} N \\ \varphi \end{bmatrix} + \begin{bmatrix} \gamma_\parallel & -ik_y \rho_s c_s / L_n - \gamma_\parallel \\ ik_y \nu_{in} v_0 / k_\perp^2 \rho_s c_s - \gamma_\parallel / k_\perp^2 \rho_s^2 & \nu_{in} + \gamma_\parallel / k_\perp^2 \rho_s^2 \end{bmatrix} \begin{bmatrix} N \\ \varphi \end{bmatrix} = 0. \quad (5.3)$$

5.1. Normalization

Normalize scales to ρ_s , frequencies to Ω_i , and so normalize velocities to c_s .

$$\frac{\partial N}{\partial t} - \frac{\rho_s}{L_n} \frac{\partial \varphi}{\partial y} + \{\varphi, N\} - \frac{\gamma_\parallel}{\Omega_i} [\varphi - N] = 0 \quad (5.4)$$

$$\frac{\partial \nabla_\perp^2 \varphi}{\partial t} + \{\varphi, \nabla_\perp^2 \varphi\} + \frac{\nu_{in}}{\Omega_i} \nabla_\perp^2 \varphi - \frac{\nu_{in} v_0}{\Omega_i c_s} \frac{\partial N}{\partial y} - \frac{\gamma_\parallel}{\Omega_i} [\varphi - N] = 0. \quad (5.5)$$

From this we can see that the relevant parameters are $k_0 \rho_s$, ρ_s / L_n , ν_{in} / Ω_i , v_0 / c_s , $\gamma_\parallel / \Omega_i$. The first three parameters are probably small, while the last two parameters can be large (stiff). **NUMERICAL IMPLEMENTATION IN PROGRESS....**